

Chi Square & Contingency

- Categorical Data

→ Non-numeric categories or labels

→ e.g., Overwatch roles or favorite hero

- Why Analyze?

→ We want to check if there are relationships between Cat. Variables

→ e.g., Does role preference influence your favorite hero?

- Chi-Square Test

→ Chi (or χ) square test helps determine association between Cat. Variables

Contingency Tables

→ Matrix showing variable distribution

→ Visualize variable relationships

e.g., "Is someone's role related to their fav. hero?"

Role	Favorite Hero				
	Mauga	Genji	Maira	Total	
	Tank	30	10	5	45
	DPS	5	40	5	50
	Support	5	10	40	55
Total	40	60	50	150	

Favorite Hero					
	Mauga	Genji	Moira	Total	
Role	Tank	30	10	5	45
	DPS	5	40	5	50
	Support	5	10	40	55
	Total	40	60	50	150

Chi Square Test

1. Formulate hypothesis

• (H₀) Null hypothesis

There is **no relationship** between role and favorite hero, and the variables are **independent**.

• (H₁) Hypothesis

There **is a relationship** between role and favorite hero, and the variables are **dependent**.

3. Calculate χ^2 Statistic

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} \quad \begin{array}{l} O_i : \text{observed freq.} \\ E_i : \text{expected freq.} \end{array}$$

$$E_{\text{Tank, Mauga}} = \frac{45 \cdot 40}{150} = 12$$

$$E_{\text{Tank, Genji}} = \frac{60 \cdot 45}{150} = 18$$

$$E_{\text{Tank, Moira}} = \frac{50 \cdot 45}{150} = 15$$

Do this for every cell.

Tank, Mauga

$$\frac{(30-12)^2}{12} = \frac{18^2}{12} = \frac{324}{12} = 27$$

Tank, Genji

$$\frac{(10-18)^2}{18} = \frac{-8^2}{18} = \frac{64}{18} \approx 3.56$$

Tank, Moira

$$\frac{(5-15)^2}{15} = \frac{-10^2}{15} = \frac{100}{15} \approx 6.67$$

Expected Frequency Table

Favorite Hero			
	Mauga	Genji	Moira
Tank	12	18	15
DPS	13.33	20	16.67
Support	14.67	22	18.33

$$\chi^2 = 27 + 3.56 + 6.67 + 5.2 + 20 + 6.17 + 6.37 + 6.55 + 25.63$$

$$\chi^2 \approx 109.15$$

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4. Degrees of Freedom (df)

$$df = (r-1) * (c-1)$$

r : row count

c : column count

$$\begin{aligned} df &= (3-1)(3-1) \\ &= 2(2) \\ &= 4 \end{aligned}$$

• P-Value $\rightarrow$ Significance level (α)

$$\alpha = 0.05$$

Using a Chi square distribution table,
the critical value is determined to be
9.49

Our Stuff

$$\chi^2 \longrightarrow 109.15$$

$$df \longrightarrow 4$$

$$\alpha \longrightarrow 0.05$$

$$\text{Critical value} \longrightarrow 9.49$$

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* Analysis ~
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1. Is critical value $< \chi^2$? **Yes!**
2. Is the difference large? **Yes! (≈ 12 times)**

$$9.49 < 109.15$$

- Reject H_0 , because χ^2 is ≈ 12 times larger than the critical value.
- Player character preferences are not random, they are related to player role. (H_1)